Homework 3

1.

For context-free grammar G_4,

$$E \to E + T \mid T$$
$$T \to T \times F \mid F$$
$$F \to (E) \mid a$$

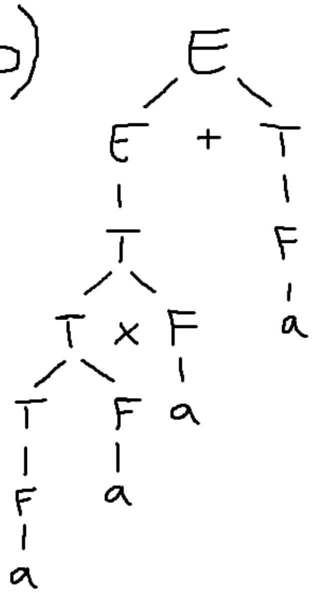
Give **BOTH** parse trees and derivations for each string:

a. a
b. $a \times a \times a + a$
c. $a + a$
d. $((a))$

a)

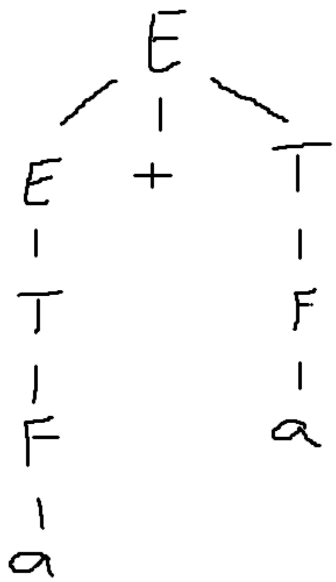
Derivation
E => T => F => a

b)



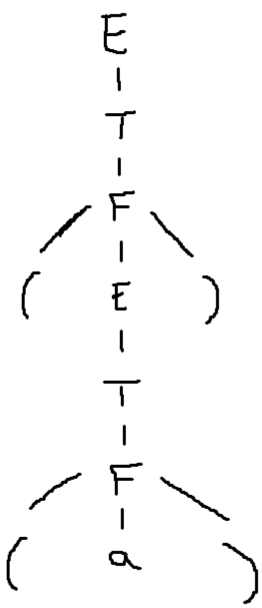
Derivation:
E
=> E + T (E → E + T)
=> T + T (E → T)
=> T × F + T (T → T × F)
=>T × F × F + T (T → T × F)
=> F × F × F + T (T → F)
=> a × F × F + T (F → a)
=> a × a × F + T (F → a)
=> a × a × a + T (F → a)
=> a × a × a + F (T → F)
=> a × a × a + a (F → a)

c)



Derivation:
E
=> E + T (E → E + T)
=> T + T (E → T)
=> F + T (T → F)
=> a + T (F → a)
=> a + F (T → F)
=> a + a (F → a)

d)



Derivation:
E
=> T (E → T)
=> F (T → F)
=> (E) (F → (E))
=> (T) (E → T)
=> (F) (T → F)
=> ((E)) (F → (E))
=> ((T)) (E → T)
=> ((F)) (T → F)
=> ((a)) (F → a)

2.

Answer each of the following questions for context-free grammar G.

$$R \to XRX \mid S$$
$$S \to aTb \mid bTa$$
$$T \to XTX \mid X \mid \varepsilon$$
$$X \to a \mid b$$

a. What are the variables of G?
b. What are the terminals of G?
c. Which is the start variable of G?
d. Give three strings in L(G).
e. True or False: $T \Rightarrow aba$.
f. True or False: $T \overset{*}{\Rightarrow} aba$.
g. True or False: $T \Rightarrow T$.
h. True or False: $T \overset{*}{\Rightarrow} T$.
i. True or False: $XXX \overset{*}{\Rightarrow} aba$.
j. True or False: $X \overset{*}{\Rightarrow} aba$.
k. True or False: $T \overset{*}{\Rightarrow} XX$.
l. True or False: $T \overset{*}{\Rightarrow} XXX$.
m. True or False: $S \overset{*}{\Rightarrow} \varepsilon$.

a. R, S, T, X
b. a, b
c. R
d. ab -> R => S => aTb => aεb = ab
   ba -> R => S => bTa => bεa = ba
   abb -> R => S => aTb => aXb => abb
e. False
f. True, T => XTX => XXX => aba
g. False
h. TRUE, T => *T in zero steps
i. True, XXX => aba (X -> a, X -> b, X -> a)
j. False
k. False
l. True, T => XTX => XXX
m. False